

Growing Up to Learn: Developmental Scaffolds for Reinforcement Learning

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Abstract

Reinforcement learning is usually framed as a choice between imitation and evaluation: copy a teacher, or learn from reward alone. Human learners live in the middle — they take instruction, keep world models, drill microtasks, practice with deliberate variation, inherit compressed experience from teachers, and do all of it inside bodies that start small. We turn five developmental intuitions into falsifiable hypotheses and test them in deliberately small systems (tabular and linear agents; gridworlds and a torque-limited pendulum) where budgets are matched to the environment step, confirmatory runs use fresh seeds disjoint from all tuning, and every primary comparison is Holm-corrected under both Mann-Whitney and Welch tests. Three core effects replicate and survive adversarial verification: (1) a learned world model supports near-sighted competence with observations removed — 97.5% blindfolded success at “home” versus 29% in a “stranger’s home” — even though its celebrated sample-efficiency advantage is fully explained by update count; (2) drill curricula over start states reach whole-task mastery 2.7x faster for epsilon-greedy learners, an effect driven by targeted structure (uniform start diversity is worthless) and bounded by exploration regime (optimistic initialization reproduces it); (3) the Shea–Morgan contextual-interference crossover appears in silico — interleaved practice loses the practice room and wins retention and transfer — and vanishes when shared features are ablated, establishing shared-parameter interference as the mechanism. A fourth result is new, to our knowledge: iterated distillation through an episodic-memory bottleneck, from plasticity-decaying teachers to fresh students, consolidates rare peak experiences that no dose-matched control can (weight copying, random optimistic advice, longer lives), while a global-optimism control marks the boundary where teaching is unnecessary. Fifth, growing bodies: on the confound-free pairing, a growth curriculum reaches adult competence with 9–42x less cumulative fall damage and 30–40% fewer steps — while two folk sub-claims (gradual growth, balance-before-walking) are significantly reversed. We report the failures alongside the successes; the surviving thesis is that developmental scaffolds work by curating which experience a learner gets, and that they pay precisely where exploration is undirected and mistakes are expensive — the condition animals actually live under.

1. Introduction

An agent trained by pure evaluation must rediscover, within a single lifetime, regularities that every previous lifetime already paid for. An agent trained by pure imitation inherits those regularities but is ceilinged by its teacher and stranded the moment the world drifts from the demonstrations. Humans occupy the productive middle: we take *instruction* — compressed generational experience — and cash it out through our own trial and error.

This paper takes five specific intuitions about that middle ground, states them as falsifiable hypotheses, and tests them end-to-end:

H1 (The Blindfold Test). Close your eyes in your own home and you can still find the fridge; close them in a stranger’s home and you are lost immediately. The motor system did not change — the *map* did. We test whether a learned world model supports competent blind action in a familiar environment and fails in an unfamiliar one, and whether the model’s celebrated sample-efficiency benefit survives an update-matched control.

H2 (Microtasks). Nobody learns soccer by only playing soccer, and no piano teacher assigns only full run-throughs. We test whether a curriculum that changes *nothing but the start-state distribution* — shooting drills, then dribbling drills, then full games — beats whole-game practice at matched budgets, and against two controls that decompose the mechanism: uniform random starts (diversity without structure) and optimistic initialization (a better explorer with no curriculum at all).

H3 (Variation Practice). Practicing a passage identically every time feels efficient and is not. Motor-learning research calls it the contextual-interference effect: blocked practice wins acquisition and loses retention and transfer. We test whether the crossover emerges in a minimal learner, and — by ablating shared features — *why*.

H4 (Generational Teaching). A teacher does not transplant their brain into a student; they transmit a curated sliver — the passages that matter, the one time they reached the summit. We test whether a lineage of short-lived, plasticity-decaying agents, each distilling its best episodes through a narrow bottleneck into a fresh student, outperforms weight copying, longer single lives, and dose-matched optimism controls.

H5 (Growing Bodies). Toddlers fall constantly and cheaply; by the square-cube law small bodies are also relatively strong. We test whether starting small and growing — paying for mistakes at toddler prices — reaches adult competence cheaper and no slower than training the adult body directly, and whether the folk staging (balance first, grow gradually) actually helps.

Our experimental philosophy is deliberate miniaturization. Every environment is small enough that budgets can be matched to the individual step, every comparison run across 30–60 fresh seeds, and every mechanism exposed by ablation. The claims are correspondingly scoped: these are existence proofs and mechanism demonstrations in minimal systems, not benchmark results. The program was subjected to an adversarial verification pass (fifteen independent audits, three lenses per experiment) followed by a pre-registered hardening round: every hypothesis re-run on fresh seeds with the auditors’ mandated controls. What failed is reported next to what survived. Notably, several failures are the most informative results in the paper.

Related Work

Each of our five experiments sits on an established literature: the spectrum of channels between imitation and pure reinforcement (EXP4’s advice bottleneck and EXP2’s boundary condition), latent learning and world models (EXP1), curricula and part-whole training (EXP2), variability of practice and contextual interference (EXP3), generational transmission (EXP4), and morphological development (EXP5). We review each thread, then locate our contribution: mostly deliberate replication-in-miniature, plus four protocols we believe are new.

Between imitation and evaluation

Static demonstrations cannot be trusted at states the teacher never visited: behavioral cloning with per-step error ϵ incurs up to $O(T^2 \epsilon)$ regret as the learner’s own mistakes shift the state distribution, and DAGger restores $O(T \epsilon)$ -order guarantees by reducing interactive imitation to no-regret online learning (Ross et al. 2011). Folding demonstrations into value-based RL buys a head start but can anchor: DQfD beat prioritized dueling DQN on 41 of 42 Atari games in the first million steps, yet its persistent margin loss left it above its own demonstrator on only 14 (Hester et al. 2018). Annealing the teacher away removes the tether — kickstarted students reach teacher level roughly 10x faster and end 42% above the from-scratch baseline precisely because the distillation weight is driven to zero (Schmitt et al. 2018) — and Jump-Start RL moves instruction entirely into the exploration distribution: a guide policy rolls in for a receding horizon while the student’s objective stays pure RL (Uchendu et al. 2023). The offline-to-online literature shows why the handoff is delicate: distribution shift at the handoff can destroy a good initial policy (Lee et al. 2021), conservative pretraining produces a characteristic dip that value calibration removes (Nakamoto et al. 2023), while simply mixing offline data into the online buffer is provably efficient under coverage (Song et al. 2023). Teacher quality caps the value of imitation, not of RL (Kumar et al. 2022),

and RLHF chains the stages — supervised imitation, then optimization against a learned proxy under a KL tether (Ouyang et al. 2022).

Theory bounds when any of this helps. A guide policy covering the optimal policy’s states turns the exponential-in-horizon sample complexity of naive epsilon-greedy sparse-reward exploration into a polynomial one (Uchendu et al. 2023), but the matching lower bound of Xie et al. (2021) shows that in the worst case the optimal algorithm either reduces purely to the reference policy or ignores it: instruction is worth exponential factors to naive explorers and approximately nothing to optimistic ones. That asymmetry is load-bearing here — it predicts the boundary result of EXP2, where the drill curriculum’s advantage disappears against an optimistically initialized whole-game learner. The channel matters independently of the content: priming values or policies from a demonstration gave no speedup where priming a dynamics model enabled one-trial pole balancing (Schaal 1997); Q-value initialization is update-for-update equivalent to potential-based shaping (Wiewiora 2003), itself the only reward modification guaranteed policy-invariant (Ng et al. 1999); and a fixed advice budget is best spent at high-importance states rather than early (Torrey & Taylor 2013). The organizing lesson we take: acceleration comes from an artifact reshaping the exploration distribution, lock-in from an artifact persisting in the objective. EXP4’s advice prime — a value artifact with no persistent loss term, freely overwritten by TD updates — sits on the benign side of that split by design.

World models and latent learning

Latent learning is the original evidence that agents acquire structure without reward: rats that explored mazes unrewarded collapsed their error curves within a day or two of the first reward to the level of always-rewarded controls (Blodgett 1929; Tolman & Honzik 1930), which Tolman (1948) synthesized into the cognitive-map hypothesis. The cautionary half of that history matters equally: the famous sunburst shortcut probe (Tolman et al. 1946) replicates poorly, exceeding chance in only 17% of 47 experiments in a recent meta-analysis (Duvelle & Grieves 2026), so we rest conclusions on repeated probes over many seeds, never one dramatic display. Dead reckoning — the competence our blindfold test isolates — is biologically real and lawful: gerbil mothers retrieve displaced pups in total darkness on direct vectors from self-motion cues (Mittelstaedt & Mittelstaedt 1980); blind and blindfolded-sighted humans path-integrate equally well, with error growing in path length and complexity (Loomis et al. 1993); and mammalian path integration accumulates drift that landmark fixes reset (Etienne & Jeffery 2004) — the pattern our slip noise, distance effects, and bump-driven belief re-anchoring are built to reproduce.

In RL, Dyna made planning “RL applied to simulated experience” (Sutton 1990), and world models scaled the idea: policies trained inside a learned dream transfer to reality, though at low dream temperature the policy exploited model flaws, scoring ~ 2086 in the dream but ~ 193 in reality (Ha & Schmidhuber 2018); one-step model errors compound with rollout horizon, which many short branched rollouts mitigate (Janner et al. 2019); latent models predicting only reward, value, and policy plan at superhuman level (Schrittwieser et al. 2020); and one fixed Dreamer configuration now spans 150+ tasks (Hafner et al. 2023). Two findings discipline our claims. Blind navigation success alone does not require an explicit model — agents sensing only egomotion reach $\sim 95\%$ success and spontaneously develop decodable metric maps (Wijmans et al. 2023) — so our inference rides on the home-versus-stranger and touch/no-touch contrasts, not on blind success per se. And experience replay is effectively a non-parametric model, so claimed model-based sample-efficiency gains can be pure update-count artifacts (van Hasselt et al. 2019) — the reason EXP1 adds an update-matched replay baseline before crediting Dyna’s speedup to the model.

Curricula and part-whole practice

Curriculum learning entered ML as easy-examples-first (Bengio et al. 2009); curriculum RL was formalized as task generation, sequencing, and transfer (Narvekar et al. 2020) using the transfer metrics of Taylor & Stone (2009). That framework’s sharpest lesson is budget accounting: most surveyed work demonstrates only “weak transfer,” with drill time treated as sunk cost. The earliest RL curriculum already won under strong accounting — easy-pole-first balancing took 73 total failures against 119 direct (Selfridge et al. 1985) — as did the direct ancestors of our soccer drills: vision-based shooting learned by starting episodes near

the goal and moving them back (Asada et al. 1996), and auto-generated shoot and dribble microtasks in Half Field Offense that beat from-scratch learning with curves offset by all source-task time (Narvekar et al. 2016). Start states alone are a powerful curriculum: reverse curricula grow the start distribution backward from the goal (Florensa et al. 2017), with theory supplied by restart distributions that cover a good policy’s state visitation (Kakade & Langford 2002); hindsight relabeling is the main non-curriculum alternative for sparse goals (Andrychowicz et al. 2017). EXP2’s drills change only the start-state distribution inside the same MDP, so we attribute any win to this restart mechanism — not to “task decomposition” as such — and charge every drill step to the shared budget.

The human part-task literature conditions the prediction rather than guaranteeing it: part training is most effective when subtask complexity is high and inter-subtask organization low (Naylor & Briggs 1963); the segmentation wins used backward chaining, training the final segment first (Wightman & Lintern 1985), yet forward chaining beat backward on a keyboard skill (Ash & Holding 1990), so our shoot-first ordering is one point in a task-dependent space. Curricula also have a budget regime: across thousands of orderings, curriculum conferred no benefit in standard supervised settings, helping only under limited budget or label noise (Wu et al. 2021) — so we separate time-to-threshold claims from asymptotic ones and scope the effect to the tested budget.

Variable practice and contextual interference

Shea & Morgan (1979) is EXP3’s target result: blocked practice looked better during acquisition (1.32 vs 1.69 s), then the ordering reversed at retention (1.31 vs 1.73 s favoring random acquisition) and transfer, with the blocked deficit largest under changed test contexts. Schema theory predicted the related variability-of-practice effect (Schmidt 1975), most strikingly when varied practice that never included the criterion distance beat practice at the criterion itself (Kerr & Booth 1978); the umbrella concept is desirable difficulties — conditions that impair acquisition while enhancing retention and transfer (Bjork 1994). The boundary conditions sit exactly where our task lives: contextual interference is most reliable when variants require different motor programs and weaker for parameter-only variation (Magill & Hall 1990); meta-analytically it is large in laboratory tasks (SMD 0.92) yet null in applied sport (Czyz et al. 2024); and in the closest analog of our paradigm — piano-like sequences practiced under tempo variability — lower variability and non-random schedules transferred better, reversing the textbook prediction (Caramiaux et al. 2018). We therefore pre-committed to publishing whichever direction the in-silico effect takes.

Across ~80 years of motor learning, categorization, perception, language, and ML the synthesis is consistent: low-variability input is learned fast and generalizes poorly; high-variability input, the reverse (Raviv et al. 2022). RL operationalizes practice variation as environment diversity: domain randomization transfers from non-realistic simulation to the real world (Tobin et al. 2017), CoinRun and Procgen exhibit an explicit train-worse/test-better crossover as level diversity grows, with severe memorization below thousands of variants (Cobbe et al. 2019; Cobbe et al. 2020), and small-scale agents memorize fixed configurations by default (Zhang et al. 2018). What that literature does not do — and EXP3 does — is hold the variant set fixed and manipulate only the practice schedule, over an engineered shared-feature substrate for interference, with the full acquisition/retention/transfer battery.

Generational transmission

Distillation lineages show students exceeding teachers: policy-distillation students match or beat their teachers at a fraction of the size (Rusu et al. 2016), and born-again networks — equal-capacity students trained on teacher outputs, generation after generation — outperform their teachers, with gains arriving in the first few generations and then saturating (Furlanello et al. 2018); self-distillation theory predicts that continued rounds eventually underfit, an inverted-U over generations (Mobahi et al. 2020). Reincarnating RL formalizes reuse of prior computation but transfers the full teacher, with no curated bottleneck and no teacher aging (Agarwal et al. 2022); distilling into a freshly initialized network within one lineage repairs damage from early non-stationarity (Igl et al. 2021); and imitating one’s own top-return episodes helps within a single lifetime (Oh et al. 2018) — a confound any generational claim must separate from the fresh-student step.

The iterated-learning tradition supplies the theory of transmission itself: human diffusion chains learning artificial languages through a bottleneck become more learnable and more structured without anyone intending it (Kirby et al. 2008); for posterior-sampling learners the chain converges to the learner’s prior, bottleneck size setting the rate, not the endpoint (Griffiths & Kalish 2007); and structure requires joint pressure for compressibility and expressivity — transmission-only chains collapse to degenerate systems (Kirby et al. 2015). In our chains the student’s post-imitation reward-driven lifetime is that expressivity pressure. Cultural transmission has reached deep RL as a single expert-to-novice step acquired zero-shot (Bhoopchand et al. 2023) and as generational accumulation that beats one long life at matched cumulative experience (Cook et al. 2024); human chains given two lives per generation accumulate by passing distilled written messages (Tessler et al. 2021) — the nearest relative of our 100-pair advice artifact. The null model is structural drift: resampling chains can wander into degenerate absorbing states, so cumulative improvement is not guaranteed (Crutchfield & Whalen 2010).

The aging half of EXP4 is motivated by plasticity loss: deep continual learners progressively lose the ability to learn at all (Dohare et al. 2024). But the same literature raises the identifiability question our controls exist to answer: periodically resetting network parameters while keeping the replay buffer already improves deep RL by curing primacy bias (Nikishin et al. 2022), so an apparent generational benefit could be “just resets” — restored plasticity rather than transmitted content. EXP4’s battery targets exactly this: no-inheritance lineages (fresh agents, no advice) isolate the reset effect alone; weight-copy separates inherited content from inherited freshness; and one-long-life controls with matched and slowed plasticity decay price the cost of aging. We present teacher aging as a model inspired by deep-network plasticity loss, not a phenomenon intrinsic to tabular learners — which is precisely what keeps the testbed clean.

Morphological development

Bongard (2011) found that robots whose body plans changed during evolution found successful controllers faster and ended more robust — but by evolutionary search over controllers, not within-lifetime RL, which is exactly the transfer our experiment tests. Developmental robotics articulates the rationale of immaturity as a dimensionality-reducing scaffold (Lungarella et al. 2003), with the complication that a single monotonic freeze-then-free schedule can be insufficient (Berthouze & Lungarella 2004). The empirical record is honestly mixed: across five morphological-development strategies on a bipedal walker, development helped in some setups and hurt in others (Naya-Varela et al. 2023), while the strongest recent positive result required realistic child-to-adult anthropometry — uniformly scaled morphology failed (Badie et al. 2025). Morphology causally modulates learnability (Gupta et al. 2021), and transferring policies through a continuous sequence of intermediate bodies beats abrupt transfer (Liu et al. 2022) — the prior for grow-linear over grow-jump.

The infant data license our cost model: walking infants average ~2,400 steps and 17 falls per hour (Adolph et al. 2012), and falling is cheap — over 90% of spontaneous infant falls are uneventful, and measured fall energy is 18.4x lower than the same falls at adult size and speed (Han & Adolph 2021), the anchor for our $(s/s_{\text{adult}})^4$ damage law. The counter-evidence is equally real: balance knowledge can fail to transfer across postures, with infants avoiding risky gaps while sitting yet falling into them while crawling (Adolph 2000), so a balance-first null would be developmentally consistent. We also model the physics honestly: the inverted pendulum’s time constant $\sqrt{l/g}$ means the small body falls faster and is harder for any delay-limited controller (Milton et al. 2009); smallness’s true advantages are the L^4 fall cost and relative actuation authority, and we disclose that direction rather than rig it. Standing precedes walking in the WHO milestone windows (WHO Multicentre Growth Reference Study Group 2006), the empirical basis for balance-first; and assistive forces annealed to zero (Yu et al. 2018) are the established non-morphological easy-start against which “growing the body specifically helps” would ultimately have to be tested.

Positioning

Much of this program is deliberate replication: preregistered developmental-psychology and motor-learning effects — latent learning, part-task curricula, the Shea & Morgan crossover, cheap infant falls — reproduced in minimal, fully inspectable RL systems where every value table, model, belief, and advice set can be printed and audited. Against that baseline we claim four protocols as new. (a) The blindfold evaluation of a

learned world model: prior work trains blind agents end-to-end (Wijmans et al. 2023) or evaluates models by dream-training transfer (Ha & Schmidhuber 2018); using an already-learned Dyna model as the substrate for belief-filtered dead reckoning, with home-versus-stranger and touch/no-touch contrasts as the discriminating controls, is a new evaluation protocol rather than a new algorithm. (b) Iterated distillation through an episodic-memory bottleneck with plasticity-decaying teachers: the pieces exist separately — generational accumulation (Cook et al. 2024), zero-shot cultural transmission (Bhoopchand et al. 2023), full-policy reuse (Agarwal et al. 2022), born-again distillation (Furlanello et al. 2018), fresh-network distillation within a lineage (Igl et al. 2021), resets whose transmission channel is the entire unfiltered buffer (Nikishin et al. 2022), and the bottleneck theory of Kirby et al. (2008) and Griffiths & Kalish (2007) — but to the best of our review no published work combines an aging teacher, fresh tabular students, and a hard countable trajectory cap with the weight-copy / long-life / no-inheritance causal battery. (c) In-silico contextual interference via engineered shared-feature interference: the RL generalization literature varies environment diversity and measures gaps (Cobbe et al. 2019) but does not run the Shea & Morgan schedule manipulation over a fixed variant set with a mechanism hypothesis and an acquisition-retention-transfer battery. (d) Morphological growth curricula with square-cube-honest physics: prior art is evolutionary (Bongard 2011), neuroevolutionary with mixed results (Naya-Varela et al. 2023), or deep and muscle-actuated (Badie et al. 2025); none runs within-lifetime tabular RL with a disclosed $\sqrt{1/g}$ difficulty direction, L^4 fall-cost scaling anchored to Han & Adolph (2021), and cumulative damage as the headline metric. Where we replicate, we say so loudly; where we claim novelty, we claim exactly these four protocols and no more.

3. Methods

3.1 Shared protocol

All agents are tabular Q-learners or linear-function-approximation Q-learners implemented in numpy; all environments are custom and fully specified in the repository. Statistical protocol, applied uniformly: interquartile-mean (IQM) point estimates; 95% percentile-bootstrap confidence intervals (10,000 resamples); two-sided Mann-Whitney *and* Welch t tests for every primary comparison, Holm-corrected within each experiment’s primary family; censored time-to-threshold values assigned budget+1 (conservative) with censoring fractions always reported. Training budgets are matched to the environment step across conditions — drill steps, practice steps, and growth-phase steps all count. Evaluation is greedy, identical across conditions, uses rng streams disjoint from training, and consumes no training budget. Hyperparameters were selected on seed ranges disjoint from the confirmatory runs (a v1 violation of this rule, caught by the audit, is disclosed in §5); all confirmatory results below are from fresh seed ranges (100+). Design amendments between v1 and v2 are versioned in the repository’s DESIGN.md.

3.2 EXP1 — The Blindfold Test

Environment. GridHome: two 13x11 apartments (HOME_A, HOME_B) with identical shells, identical start (“bed”) and goal (“fridge”) coordinates, and different interior walls. Motor noise: 10% probability the executed move is perpendicular to the intent. Reward 1 at the fridge; episode cap 60 steps (~3x shortest path); gamma 0.97.

Agents. Q-learning; Dyna-Q with 20 planning updates per real step (its transition/reward model doubles as the agent’s imagination); and — mandated by the audit — ReplayQ, an update-matched control performing 20 uniformly-replayed updates per real step from its own transition buffer, with no model.

Blindfold protocol. After 40k training steps in HOME_A, observations are removed. The agent dead-reckons: a belief over states, initialized at the known start, advanced through the learned transition model by each executed action; actions chosen greedily by belief-weighted Q. The *touch* variant additionally filters the belief by the learned bump likelihood — a hand trailing the wall. Conditions: sighted-A, blind-A, blind-A-touch, blind-B, blind-B-touch (both using HOME_A’s model and values inside HOME_B), sighted-B-transfer (HOME_A’s values with full observation in HOME_B), and matched random floors in both homes. 30 seeds.

3.3 EXP2 — Microtasks

Environment. SoccerGrid: an 11x7 pitch; the ball is carried, loose at a cell, or scored; 4 moves + shoot; shots score with probability $\max(0, 1 - d/5)$ at Chebyshev distance d from goal center, and a miss ends the episode; reward only on goals; cap 100 steps.

Conditions (60k steps each, one shared Q-table per agent, 30 seeds): **whole** (full games from kickoff); **drills-varied** (20% of budget spawning with the ball at random attacking-third cells, 20% spawning at random own-half cells with the ball at center, then full games — a reverse curriculum over start states with no reward changes); **drills-fixed** (same phases, single fixed spawn each); **explore-starts** (uniformly random agent and ball positions for 40% of budget — diversity without structure); **whole-optimistic** (full games, $Q_0 = 1.0$ — a better explorer, no curriculum).

3.4 EXP3 — Variation Practice

Environment. PianoPiece: three 12-note passages assembled from a shared library of six 3-note motifs, one exception position per passage; 8 keys; +1 per correct note; score = fraction correct.

Agent. Linear $Q(s,a) = w_a \cdot \phi(s)$, with $\phi = \text{onehot}(\text{motif, position-in-motif}) ++ \text{onehot}(\text{passage}) ++ \text{onehot}(\text{position})$. The shared motif slots are where interference can live. The mechanism-control agent replaces ϕ with passage-local features only (no shared slots).

Conditions (450 episodes each, 40 seeds): **blocked** (one passage per third of budget, block order counterbalanced across seeds — seed % 6 selects one of the six orders, removing the v1 recency confound); **interleaved** (uniform random passage each episode); both feature maps crossed with both schedules for the mechanism control. Test phase (learning off): retention on all three passages; transfer to a novel passage built from the same motifs.

3.5 EXP4 — Generational Teaching

Environment. TrapGrid (15x11): start at the left; reward 10 far right; three candy cells (reward 0.3, terminal) near the start that end episodes early and throttle deep exploration; cap 120 steps; deterministic moves.

Lifetimes. One life = 15k steps with plasticity decay: $\text{lr}(\text{age}) = 0.3 \cdot 2^{-(\text{age}/5000)}$, $\text{eps}(\text{age}) = 0.4 \cdot 2^{-(\text{age}/5000)}$. Old agents are rigid. Each agent keeps an episodic memory of its top-3 episodes by return.

The bottleneck. At death, a teacher emits advice: the deduplicated (s,a) pairs of its best-remembered episodes, capped at 100. A fresh student is primed $Q[s, a_{\text{advice}}] = 5.0$, then lives normally.

Conditions (5 generations x 15k = 75k steps each, 60 seeds): **generational-distill** (and a tie-break robustness variant); **weight-copy** (student inherits the full Q-table, fresh schedules); **one-long-life** and **one-long-life-slow** (single agents, two decay rates); **no-inheritance** (five independent lives); and the audit-mandated optimism-decomposition controls: **random-advice** (100 *random* state-action pairs primed to 5.0 each generation — dose-matched enthusiasm, no content), **optimistic-init** (one long life, $Q_0 = 5.0$ everywhere), and **constant-eps-life** (exploration never decays — tests whether plasticity decay alone strands the long life).

3.6 EXP5 — Growing Bodies

Environment. BalanceBot: a torque-limited inverted pendulum with body scale s : length s , mass $15s^3$, torque limit $40s^2$ (square-cube: small bodies are relatively stronger but dynamically faster per fixed control step — the per-step difficulty is not rigged in either direction). Falls ($|\theta| > \pi/5$) end the episode and cost damage $(s)^4$ — impact energy. Torque noise 5% of the limit. “Walking” = tracking a lean target that switches every 40 steps; evaluation is always on the adult body ($s = 1.0$) with walking targets. 765 discretized states, 5 torque actions; 120k steps; 40 fresh seeds.

Conditions. `adult-walk`; `adult-balance-first` (targets pinned to zero for the first 30%); `grow-linear` and `grow-adaptive` (size 0.5 $\rightarrow$ 1.0 on a schedule, or +0.05 whenever the recent no-fall rate exceeds 70%, both balance-first); `grow-jump` (0.5 until 60%, then 1.0); `grow-linear-walk` and `grow-adaptive-walk` (growth with walking from the start — the audit-mandated factorial cells that isolate morphology from task staging). Robustness arms rerun the primary pairing under a muscle-torque s^3 law and under size-scaled damping.

4. Results

Every claim below is scored by its pre-registered rule; “supported” and “refuted” refer to those rules, evaluated on fresh confirmatory seeds with Holm-corrected Mann-Whitney and Welch tests. Figures 1–5 show the primary curves; the interactive version of every figure, with per-seed data, is at the project site.

4.1 EXP1: the map is for acting blind, not for learning fast

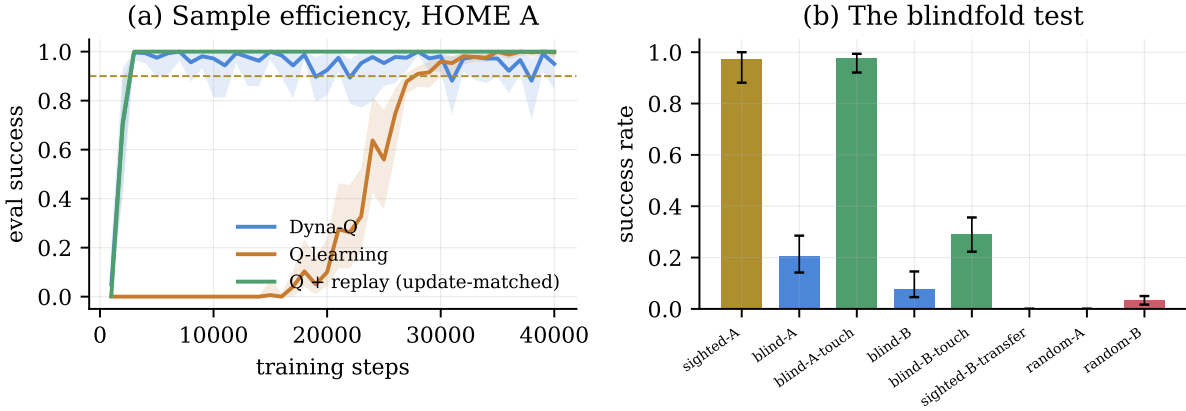


Figure 1: EXP1. (a) Sample efficiency in HOME_A: the update-matched replay control (no model) matches Dyna-Q exactly; the speedup is update count. (b) The blindfold test: touch-filtered blind navigation matches sighted at home and collapses in the stranger’s home.

The blindfold results (Fig. 1b). Sighted competence in HOME_A is 0.973 (IQM success). Pure dead reckoning collapses to 0.204 — the v1 prediction that an internal model alone sustains blind competence under motor noise is **refuted**; compounding slip drift destroys the open-loop belief, exactly as the imitation-learning literature’s compounding-error analyses would predict. But dead reckoning *plus a learned touch filter* — belief updates multiplied by the bump likelihood, a hand on the wall — reaches **0.975, statistically indistinguishable from sighted** (0.975 vs 0.973). The same machinery transported to the stranger’s home collapses to 0.290 against a measured matched random floor of 0.033 (blind-A-touch vs blind-B-touch: Welch Holm $p = 6.7e-18$; the symmetric touch-pair comparison was promoted to primary by the audit after the pure-pair comparison proved knife-edge). Sighted transfer with HOME_A’s values fails entirely in HOME_B (0.000): the bottleneck is layout knowledge, not vision.

The sample-efficiency twist (Fig. 1a). Dyna-Q reaches 90% success in 2,562 steps versus 25,500 for Q-learning (Welch Holm $p = 7.3e-31$) — the classic result. The update-matched ReplayQ control lands at **2,500 steps**, closing 100% of the gap (Dyna vs replay: $p = 0.55$). The strong claim “the model causes the speedup” is **refuted**: at matched update counts the advantage is fully an update-count effect (van Hasselt et al.’s warning, confirmed in the smallest possible setting). What replay cannot do is walk blind — no buffer of past transitions yields a belief update for a novel action sequence. The world model’s non-redundant value in this system is counterfactual use, not sample efficiency.

4.2 EXP2: drills work — for the learner that needs them

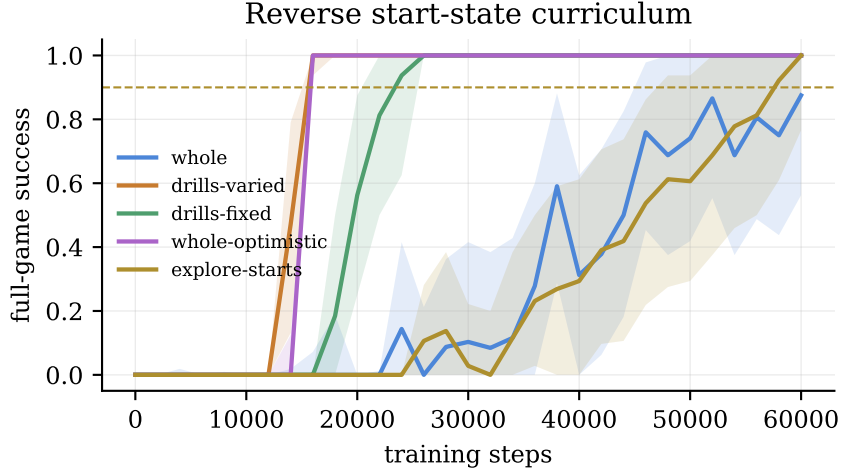


Figure 2: EXP2. Full-game success from kickoff. Drilled start-state curricula lift off during the drill phases; uniform exploring starts do nothing; optimistic initialization needs no curriculum.

On fresh seeds, **drills-varied** reaches 90% full-game success in **15,125** steps (IQM) versus **40,375** for **whole** (MW Holm $p = 3.9\text{e-}9$, Welch Holm $p = 3.4\text{e-}9$; 0/30 vs 6/30 seeds censored): a 2.7x speedup from changing nothing but where episodes start. The two controls decompose the mechanism. **explore-starts** — the same 40% of budget spent on uniformly random starts — helps not at all (43,375; $p = 0.39$ vs whole): start-state *diversity* is worthless; the drills’ *targeted* structure (starts concentrated where value already flows) carries the entire effect (drills vs explore-starts: Welch Holm $p = 4.9\text{e-}14$). **whole-optimistic** reaches 16,000 steps with no curriculum at all — statistically indistinguishable from the drilled learner ($p = 0.18$) — so the drill benefit is an **exploration effect, bounded by exploration regime**: it is large for undirected (epsilon-greedy) learners and unnecessary for optimistic ones, precisely the boundary the guide-policy theory predicts.

4.3 EXP3: the crossover replicates, and the mechanism is proven

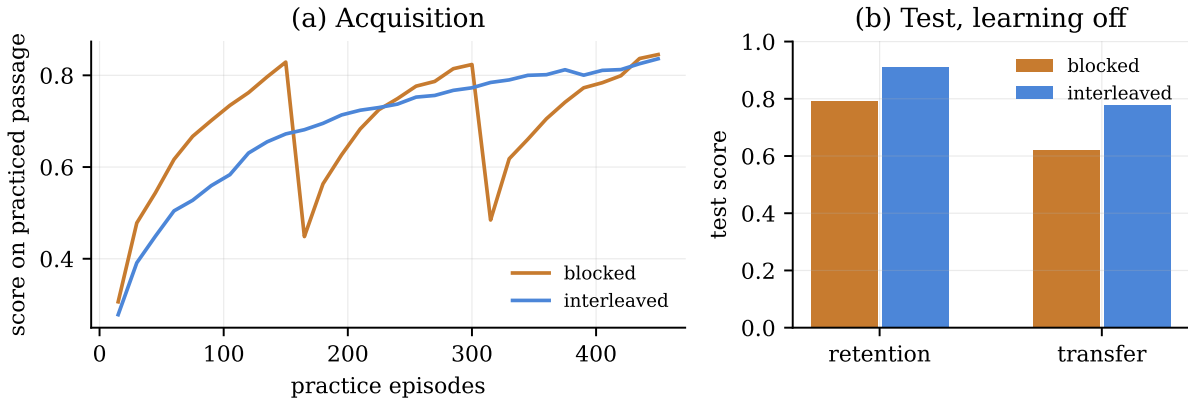


Figure 3: EXP3. (a) Acquisition: blocked practice leads during training. (b) Test with learning off: interleaved wins retention and transfer — the contextual-interference crossover.

With block order counterbalanced (killing the v1 recency confound), fresh-seed results: acquisition — blocked

0.796 vs interleaved 0.777, direction as predicted but no longer significant (MW $p = 0.076$): **boundary**. Retention — interleaved **0.910** vs blocked 0.793 (Welch Holm $p = 5.1e-10$): **supported**. Transfer to a novel passage — interleaved **0.779** vs blocked 0.621 (Welch Holm $p = 4.7e-6$): **supported**. The order-relative analysis shows the retention gap lives on earlier-drilled passages, as shared-weight drift predicts, and the v1 “blocked wins the just-drilled passage” reversal shrinks to noise once orders are counterbalanced — it was an order artifact.

The mechanism control is the sharpest result: rebuild the learner with passage-local features (no shared motif slots) and the entire crossover **vanishes** — retention gap $p = 0.92$, transfer at chance for both schedules. Contextual interference here is not a story about attention or effort; it is shared-parameter interference, demonstrated by ablation.

4.4 EXP4: the bottleneck beats every dose-matched control

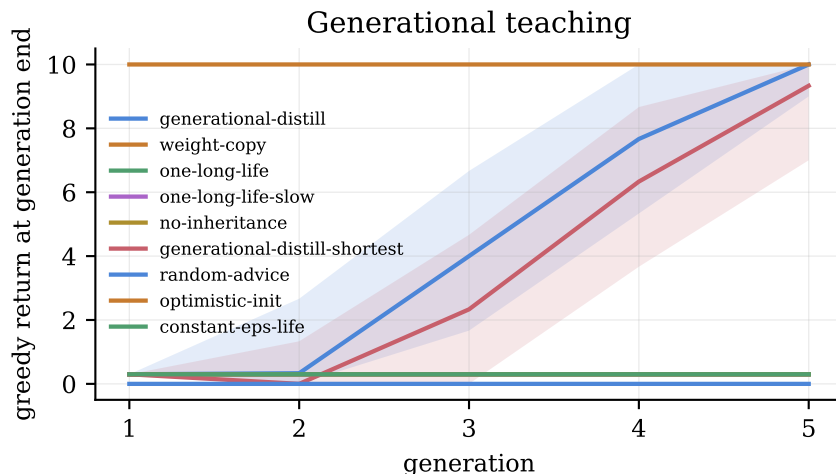


Figure 4: EXP4. Greedy return at each generation’s end. The distill lineage ratchets to the distant goal; every dose-matched control stays at candy; random advice poisons below the floor.

At 60 fresh seeds: **generational-distill** ends at IQM **10.0** (48/60 lineages consolidate the distant goal; mean 8.0). Every v1 baseline — weight-copy, both long lives, no-inheritance — ends at exactly **0.3 on every seed** (candy; all four comparisons Welch Holm $p = 1.2e-20$). The mechanism telemetry shows why: long-lived agents *remember* the mountain (best-episode return ~ 10) while *doing* candy for 75,000 steps — knowledge they can no longer consolidate. The bottleneck converts exactly that gap into the next generation’s prior.

The audit-mandated controls decompose the effect. **random-advice** — the same dose of optimism on random state-action pairs — ends at **0.0**, *below* the candy floor (scattered optimistic pins poison the greedy policy into loops): advice works because of its content, not its enthusiasm (Welch Holm $p = 2.0e-20$). **constant-eps-life** (exploration never decays) still ends at 0.3 on all 60 seeds — undecayed epsilon alone does not rescue the long life, so the registered “plasticity decay is the strander” rescue-claim is **refuted** as stated (the lr component still decayed; the triangulation is in the repository). And **optimistic-init** — global $Q_0 = 5.0$, one long life — solves TrapGrid outright, 60/60, beating even the distill lineage (Welch Holm $p = 6.0e-4$). The strong conjunctive claim “advice content matters beyond any optimism control” is therefore **refuted**, and the honest statement of H4 is conditional: *among content-bearing channels and dose-matched controls, episodic distillation is uniquely effective; where blanket optimism over the whole state space is feasible, no teacher is needed*. Blanket optimism is feasible in a 165-state gridworld; it is not a scalable strategy where states are vast and optimism is expensive to burn off — which is where teaching should matter, a prediction this miniature cannot itself test.

4.5 EXP5: growth is cheaper and faster; the folk stagings reverse

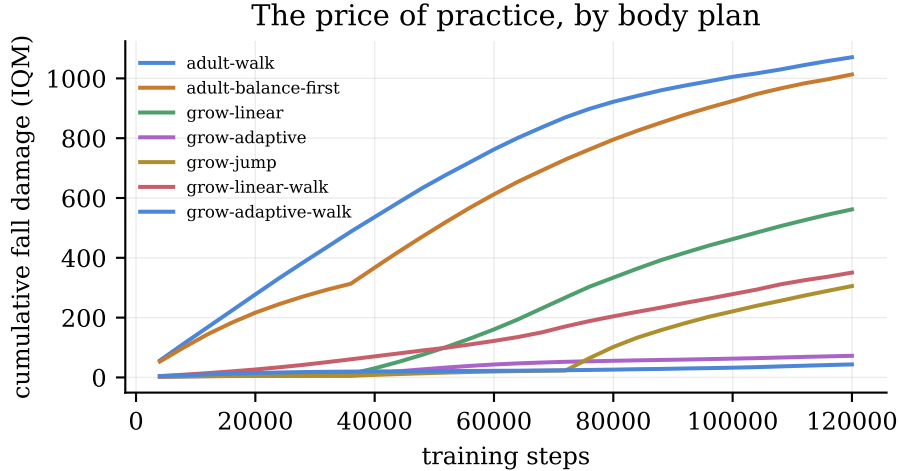


Figure 5: EXP5. Cumulative fall damage during training, by body plan. Growth curricula reach adult competence at a fraction of the damage.

On the audit-mandated confound-free pairing (both arms walking from the start): **grow-linear-walk** reaches adult competence with damage IQM **100.4** vs **928.5** for **adult-walk** (9.2x; Welch Holm $p = 3.8e-35$); **grow-adaptive-walk** with **22.0** (42x; Welch Holm $p = 4.3e-30$) — robust to censoring treatment and to both physics robustness arms (muscle-torque s^3 law; size-scaled damping). Steps-to-competence: growth is not merely no-slower but significantly **faster** — grow-linear-walk 49,000 vs 81,000 (difference CI [-40,600, -23,800]); grow-adaptive-walk 52,000 vs 81,000 (CI [-39,800, -15,400]).

Both folk sub-claims **reverse**. Balance-first staging is slower on both matched-morphology pairings (adult: 101,600 vs 81,000, Welch Holm $p = 7.4e-4$; grow-linear: 95,800 vs 49,000, Welch Holm $p < 1e-9$): weight-shift practice visits the very states balance needs, and a balance-only phase starves the rest of the table. And gradualism reverses: one abrupt jump from half-size to adult is *cheaper* than fine-grained growth (damage 156.2 vs 446.8, Welch Holm $p = 5.1e-7$). The savings come from being small while incompetent and spending little total time mid-growth — not from smoothness of the schedule, and not from imitating infant task staging. What infant development optimizes is evidently not the value function’s learning curve alone.

5. What failed, and what the failures bought

Four registered claims were refuted and two landed on boundaries; none were quietly dropped. (1) The strong Dyna claim failed its update-matched control — the correct citation for the speedup is update count, and the model’s real contribution in this system is blind action. (2) Pure dead reckoning is not home-competent under motor noise; competence needs closed-loop touch. (3) The acquisition half of contextual interference did not reach significance once block order was counterbalanced — the practice-room advantage of blocked practice is partly recency artifact even in silico. (4) H4’s strong conjunction failed against global optimism, which converts to the paper’s sharpest scope statement: instruction pays where exploration is undirected and optimism is unaffordable. (5–6) Gradualism and balance-first reversed — folk developmental staging did not survive contact with a system where only the value function matters.

We also disclose two v1 protocol violations caught by the audit and repaired in v2: hyperparameters tuned on seed ranges overlapping confirmation (fixed by fresh-seed confirmatory reruns with disjoint ranges), and one outcome-calibrated design choice (EXP4’s episodic-memory tie-break) that is now reported under both variants (IQM 10.0 earliest-tie vs 9.33 shortest-tie; the conclusion is tie-break-robust at 60 seeds).

6. Discussion

The five results compose into one operational thesis. Every scaffold that survived works by **curating which experience the learner gets** — drills reposition it, variation shuffles it, teachers select its peaks across lifetimes, small bodies discount its price — while the learner’s update rule stays untouched and free to disagree. The two boundary results (EXP2, EXP4) add the scope condition: curation pays when exploration is undirected and optimism is expensive. That condition is not a weakness of the thesis; it is a description of animal life. A creature with real fall costs, a finite lifetime, and a state space it cannot enumerate is precisely the epsilon-greedy learner of these experiments, not the globally-optimistic one.

The blindfold result reframes what a world model is *for* in the smallest possible setting: not (only) faster credit propagation — replay does that — but action under counterfactual observation: acting on experience the agent is not currently having. The generational result, to our knowledge the first demonstration of iterated distillation through an episodic bottleneck from plasticity-decaying teachers to fresh tabular students, exhibits in miniature the ratchet that iterated-learning theory predicts: the transmission bottleneck filters consolidated mediocrity and transmits structure — here, literally the map of the one good day. And the growing-bodies result puts a number on a developmental folk claim while refuting two others: what makes starting small valuable is the price of falling, not the smoothness of growth or the staging of tasks.

7. Limitations

Everything here is deliberately miniature: tabular and linear learners, hand-built environments, one layout per world (a layout-resampling replication is future work). Effects proven by ablation in minimal systems demonstrate mechanism existence, not magnitude at scale. The optimism boundary is specific to enumerable state spaces. EXP5’s damage law and torque scaling are first-principles stylizations; two robustness arms support the ordering, but a full sensitivity sweep over the physics is open. Statistical power at 30–60 seeds resolves the large effects reported; several boundary results (EXP3 acquisition) are direction-consistent but under-powered by design honesty rather than certainty.

8. Reproducibility

Everything is public: environments, agents, experiment scripts, 245 tests, the versioned design document with amendments, all raw per-seed results, the paper source, and the interactive site. One command reruns any experiment (`python -m devrl.run_all`); every number in this paper is generated by a committed script from a committed result file. Repository: <https://github.com/andrewboldi/developmental-rl>. Interactive results: <https://andrewboldi.github.io/developmental-rl/>.

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